

GUJARAT POLICE INNOVATION CHALLENGE 2026

GUSIP

Gujarat Unified Surveillance Intelligence Platform

Hybrid architecture: **Model 1** registry/GIS + **Model 2** unified viewing + **Model 3** federation middleware.

Not Model 4 — departmental VMS is not replaced. Intelligence overlay, not a statewide NVR.

PoC ~50 cameras + 31 official Sentinel feeds · scale path 80,000+

Problem

- Gujarat CCTV is large, departmental, and mixed (police, traffic, municipal, highway, railway, coastal).
- Replacing every VMS is unaffordable and operationally unacceptable.
- The force still needs statewide watchlists, multi-camera tracking, and one control-room picture.

Design rule: centralise metadata, events, clips, and alerts — not 24×7 video.

Approach (what to tick on the form)

Official model	GUSIP
Model 1 (mandatory)	Camera registry, PostGIS, health, coverage gaps
Model 2	Unified wall, ANPR, watchlist, Investigate, cases
Model 3	RTSP / ONVIF / vendor / Sentinel adapters; VMS untouched
Model 4	Not this product

Primary solution model: Hybrid / Innovative Architecture. Confirm Model 1.

Architecture (one sentence)

Departmental VMS → read-only adapters → federation bus (Redis/Kafka) → intelligence (ANPR, track, watchlist, GIS) → React control room.

Stores: PostgreSQL+PostGIS, Redis, MinIO (encrypted snapshots). On-premise Docker Compose; Kubernetes for city/state.

See [architecture.svg](#) (upload that file for the HLD/architecture form field).

Live path the jury can run

1. Control room → **Gov feeds** → Sync Sentinel (live.sentinelgujarat.in).
2. Play an official camera through the GUSIP stream proxy.
3. Add the jury plate on Watchlist; wait for sequential ANPR sample (ffmpeg + Tesseract).
4. Own/demo wall shows federated RTSP/ONVIF/vendor cameras and scripted corridors.
5. Investigate GJ 01 ST 0001 with purpose = evaluation → GIS hops.

You may demo own data, but the form requires the provided government feeds as well.

Control-room behaviour

- Watchlist hit auto-focuses the camera (large pane), coalesces repeats ($\times N$).
- Four roles: operator, investigator, coordinator, admin.
- Search requires an investigation purpose (audited).
- Coordinator is home-department scoped; break-glass is time-boxed + extra audit.
- Operators cannot export CSV/case files.

Impact

- Time-to-alert: PoC corridor targets < 8 seconds camera → inbox.
- One GIS journey across districts instead of phone hops.
- Avoid storing ~80,000 continuous streams (vs Model 4 central VMS).
- GIS gap analysis for where to buy the next camera.
- One VMS outage does not blank the state picture.

Stack

React (Vite) · FastAPI · PostgreSQL/PostGIS · Redis · MinIO · Docker · Leaflet · Tesseract · YOLOv8 hook (simulate in PoC)

APIs: Sentinel live-feed simulator, OSM tiles, GUSIP REST/WebSocket. Cloud: on-premise.

VAHAN / SARTHI / eGujCop / AFIS are interface stubs only — no live queries without MoU and mTLS.

Ask

Fund GUSIP as **intelligence middleware + regional AI**, not as a statewide NVR.

Use this PoC to prove the event contract (own feeds + official Sentinel data), then attach real ONVIF/RTSP in one commissionerate before buying GPU at scale.

Demo: <http://localhost:8080> · operator / GUSIP@ops2026

Authorised law-enforcement evaluation only.